European Power Fair Value Prototype: DE-LU Day-Ahead

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1. Executive Summary & Market Choice

The Prototype: An end-to-end pipeline forecasting DE-LU hourly Day-Ahead (DA) prices using XGBoost. It translates 24-hour forecasts into a Baseload Fair Value to identify Week-Ahead trading opportunities.

Market Choice: DE-LU (Germany-Luxembourg). Chosen because of its high liquidity and extreme volatility. This is driven by "Renewable Cannibalization" (Wind/Solar), making it a high-alpha environment for fundamental modelling.

Target Choice: Option A (Hourly DA to Baseload). By modelling every single hour the 'Duck Curve' (midday solar surplus) and the morning demand spikes affect the price can be seen.

2. Data Engineering & Ingestion

Source: ENTSO-E Transparency API (Official Python Client).

Time-Series Alignment: UTC Synchronization: All data ingested in UTC to ensure a linear physical timeline.

- **DST Handling:** Converted to Europe/Berlin only for feature extraction. The 23/25-hour DST jumps were handled by reindexing to a continuous UTC range. Thus, preventing duplicate hour errors.

Data Quality Audit (QA):

Metric	Raw Observation	Treatment/Justification
Coverage	216.75%	Identified as overlapping API fragments; resolved via UTC deduplication.
Missingness	<1%	Linear Interpolation for Wind/Solar; Forward Fill for Prices (Market Continuity).
Outliers	5 Extreme Events	Validated against historical scarcity/negative price events; retained for tail-risk.

3. Modelling & Merit Order Physics

Feature Engineering:

- **Residual Load (Primary Driver):** Actual_Load - (Wind + Solar). Represents the net demand that must be met by marginal thermal units (Gas/Coal).
- **Cyclical Encoding:** sin(hour) and cos(hour) transforms to preserve the 23:00 to 00:00 temporal proximity.

- **Interaction Term:** Solar * Hour_Cos to explicitly model the Duck Curve (midday price crashes).
- **Autoregressive Lags:** 24h and 168h (Weekly) price lags to capture market memory.

Validation & Performance:

Method: Expanding Window Walk-Forward Validation (No look-ahead bias).

Metric	Seasonal Naive (Baseline)	XGBoost	Improvement
MAE	€30.46 /MWh	€15.11/MWh	50.4% Uplift
Tail Risk (P95)	€89.12 /MWh	€43.49 /MWh	Lower Exposure

4. Prompt Curve Translation

Methodology: The 24 hourly predictions are averaged to derive a Baseload Fair Value. This is compared against the Market Forward Price to calculate the Edge.

Current Trade Signal:

- **Model Fair Value:** €100.17 | **Market Price:** €95.00 | **Edge: +€5.17 (BUY)**
- **Desk Action:** Initiate a **Long Position** on the Week-Ahead contract. The model identifies fundamental scarcity and market tightness that the forward curve has underpriced.
- **Invalidation Logic:**The signal is voided if renewable forecasts are revised upward by >15% or if a significant industrial "Demand Side Response" event occurs, which would dampen the expected price peak.

5. AI-Accelerated Workflow

Implementation: Automated "Drivers" Commentary (ai_agent.py).

- **Process:** The model outputs (SHAP values, Edge, and Metrics) are fed into a prompted LLM via GeminiAPI.
- **Productivity Multiplier:** Automates the task of writing the Morning Note for traders.

Control & Safety: Prompt only uses internal variables. Hallucinations are prevented by strict temperature settings and a data-schema constraint.

Auditability: All prompts and raw JSON responses are logged to logs/ai_logs.txt for compliance.

6. Risk Analysis & Alpha Integrity

- **MAE (€15.11):** Baseline uncertainty cost for position sizing.
- **Tail Risk (P95: €43.49):** Model underestimation during steep "Net-Load Ramps" where the Merit Order curve is most non-linear.
- **Max Error (€153.83):** Black-swan outlier (e.g., unplanned plant outages or supply shocks).

Trading Invalidation Logic:

Signals are automatically voided if:

1. Fuel Price Shock: TTF (Gas) intraday move >5% , shifting the marginal cost floor.
2. Regime Shift: Delta between 24h and 168h lags $>3\sigma$, signaling a break in seasonality; trade size reduced to 25%.

7. Performance

Final Forecast Performance (Walk-Forward Validation)

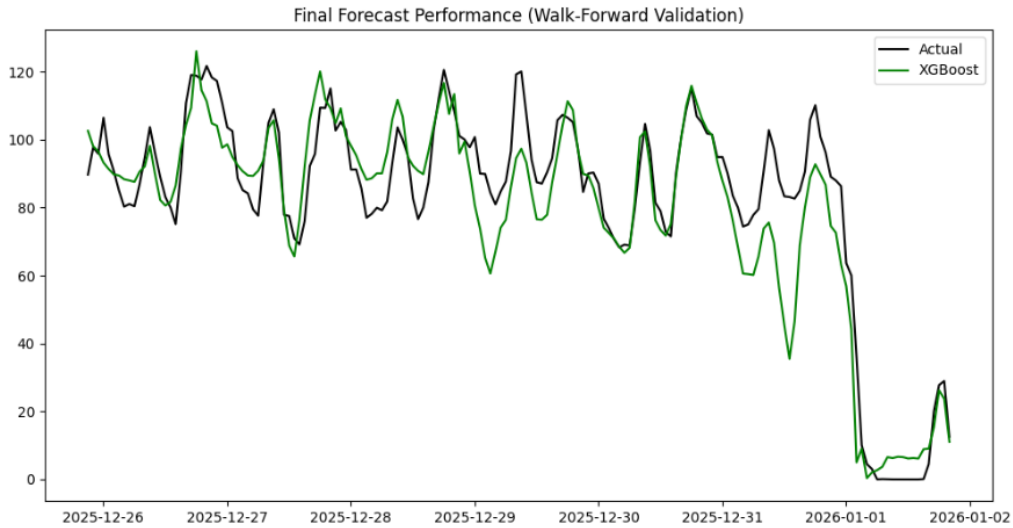

Figure 1: Time-Series Alignment of XGBoost Predictions vs. Actual DE-LU Settlement Prices.

SHAP Plot

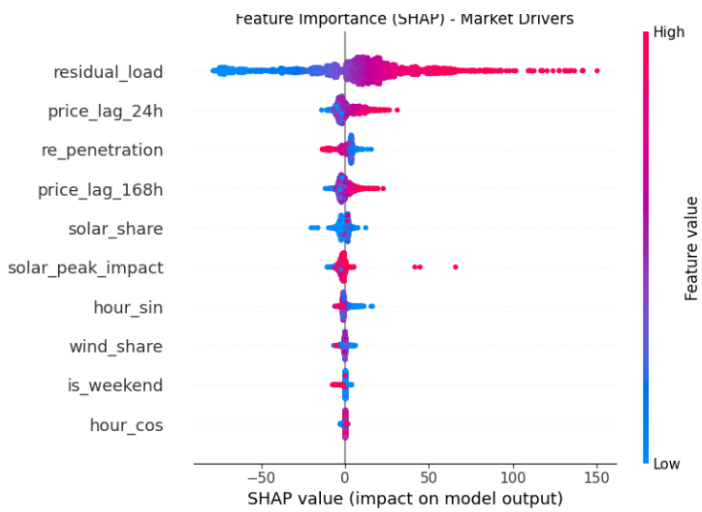

Figure 2: Shapley Additive Explanations (SHAP) - Quantifying Feature Contribution to Price.